

Engineering Physics: BAPHY105

Postulates of Quantum Mechanics

Dr. Samir Ranjan Meher

samirmeher@vit.ac.in, samirmeher@gmail.com

Department of Physics, VIT Vellore

Lecture Outcomes

- 1 Basic Postulates of Quantum Mechanics
- 2 Postulate 1: State of a System
- 3 Postulate 2: Observables and Operators
- 4 Postulate 3: Measurement Outcomes
- 5 Postulate 4: Probabilistic Outcome of Measurements
- 6 Postulate 5: Time Evolution of a System

Postulates of Quantum Mechanics

- The formalism of quantum mechanics is based on **five postulates** which are the outcome of a **range of experimental observations**.
- The prediction of these postulates has been verified by a rich collection of experiments.
- In **classical mechanics**, the state of a particle at any time t is specified by two fundamental dynamic variables: the **position $\vec{q}(t)$** and the **momentum $\vec{p}(t)$** . All other physical quantity relevant to the particle can be calculated in terms of these two dynamic variables.
- The quantum mechanical counterparts of these ideas is captured in terms of specific postulates which enable us to understand:
 - ① How a quantum state is described mathematically at a given time t ?
 - ② How to calculate the various physical quantities from this quantum state?
 - ③ How to describe the time evolution of a quantum state corresponding to a system?

Postulate 1: State of a System

- ✓ The state of any physical system at time t is specified by a state vector $|\psi(t)\rangle$ in a Hilbert space $\mathcal{H}$.
- ✓ $|\psi(t)\rangle$ contains all the needed information about the system.
- ✓ Any superposition of state vectors is also a state vector.
- ✓ The abstract state vector $|\psi(t)\rangle$ can be represented in a basis as:
 - ① Position basis $\rightarrow$ Position representation: $\psi(x, t) = \langle x | \psi(t) \rangle$
 - ② Momentum basis $\rightarrow$ Momentum representation: $\phi(p, t) = \langle p | \psi(t) \rangle$
- ✓ The wave functions must be physically acceptable:
 - The wave functions along with their first derivatives must be **continuous, single values and finite** everywhere.
 - The wave function should also be **square integrable**.

- ✓ According to Born's probabilistic interpretation, the square of the norm of $\psi(x, t)$ represents the position probability density.

$$P(x, t) = |\psi(x, t)|^2$$

- ✓ Similarly, the square of the norm of $\phi(p, t)$ represents the momentum probability density.

$$P(p, t) = |\phi(p, t)|^2$$

- ✓ The state of a system does not have to be represented by a single wave function. It can be represented by a **superposition of two or more wave functions**.

$$|\psi\rangle = \sum_i \alpha_i |\psi_i\rangle$$

- ✓ Here, α_i are complex numbers and the quantity

$$P = \left| \sum_i \alpha_i |\psi_i\rangle \right|^2 = |\alpha_i|^2 = P_1 + P_2 + P_3 + \dots = 1$$

represents the probability for this superposition if $|\psi_i\rangle$ are normalized.

- ✓ Here, P_i is the probability of finding the system in the state $|\psi_i\rangle$.

Problem Consider a system whose state is given in terms of an orthonormal set of three vectors: $|\phi_1\rangle$, $|\phi_2\rangle$ and $|\phi_3\rangle$ as,

$$|\psi\rangle = \frac{1}{\sqrt{3}}|\phi_1\rangle + \frac{2}{3}|\phi_2\rangle + \frac{\sqrt{2}}{3}|\phi_3\rangle$$

verify that $|\psi\rangle$ is normalized. Calculate the respective probabilities of finding the system in any one of the states $|\phi_1\rangle$, $|\phi_2\rangle$ and $|\phi_3\rangle$. verify that the total probability is one.

Solution:

$$\begin{aligned}\langle\psi|\psi\rangle &= \frac{1}{3}\langle\phi_1|\phi_1\rangle + \frac{4}{9}\langle\phi_2|\phi_2\rangle + \frac{2}{9}\langle\phi_3|\phi_3\rangle \\ \Rightarrow \langle\psi|\psi\rangle &= \frac{1}{3} + \frac{4}{9} + \frac{2}{9} = 1\end{aligned}$$

Therefore $|\psi\rangle$ is normalized.

Since, $|\psi\rangle$ is normalized, the probability of finding the system in $|\phi_1\rangle$ is,

$$P_1 = |\langle\phi_1|\psi\rangle|^2 = \left| \frac{1}{\sqrt{3}}\langle\phi_1|\phi_1\rangle + \frac{2}{3}\langle\phi_2|\phi_2\rangle + \frac{\sqrt{2}}{3}\langle\phi_3|\phi_3\rangle \right|^2 = \frac{1}{3}$$

Similarly,

$$P_2 = \frac{4}{9}; \quad P_3 = \frac{2}{9}$$

$$\therefore P = P_1 + P_2 + P_3 = \frac{1}{3} + \frac{4}{9} + \frac{2}{9} = 1$$

Postulate 2: Observables and Operators

- ✓ To every physically measurable quantity A , called an **observable or dynamic variable**, there corresponds a **linear Hermitian operator $\hat{A}$** whose eigenvectors form a complete basis.
- ✓ An observable is a **dynamical variable that can be measured**. The dynamical variables encountered in classical mechanics are: **position, linear momentum, angular momentum and energy**.
- ✓ Examples:
 - Position : $\hat{x} \equiv \hat{r}$
 - Momentum : $\hat{p} = -i\hbar \frac{d}{dx} \equiv -i\hbar \vec{\nabla}$
 - Energy : Hamiltonian $\hat{H} = i\hbar \frac{\partial}{\partial t}$

- ✓ if $\hat{A}$ represents any observable A ,

$$\hat{A}|\phi_n\rangle = a_n|\phi_n\rangle$$

- ✓ Here, a_n are the possible measurement outcomes for the eigenstate $|\phi_n\rangle$.
- ✓ We know that a quantum state can be written as the superposition of a number of states as,

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle$$

- ✓ The probability of measuring the eigenvalue a_n is,

$$P(a_n) = |c_n|^2$$

Postulate 3: Measurement Outcomes

- ✓ Measurement of an observable A is represented formally by the action of $\hat{A}$ on a state vector $|\psi(t)\rangle$.
- ✓ The only possible result of such measurement is one of the eigenvalues a_n (which are real) of the Hermitian operator $\hat{A}$.
- ✓ If the result of a measurement of A on a state $|\psi(t)\rangle$ is a_n , then the **system immediately after the measurement collapses to $|\phi_n\rangle$** .

$$\hat{A}|\psi(t)\rangle = a_n|\phi_n\rangle$$

- ✓ Here, $a_n = \langle\phi_n|\psi(t)\rangle$ is the component of $|\psi(t)\rangle$ when projected onto the eigenvector $|\phi_n\rangle$.

Postulate 4: Probabilistic outcome of measurements

- ✓ When measuring an observable A of a system in a state $|\psi\rangle$, the probability of obtaining one of the eigenvalues a_n corresponding to the operator $\hat{A}$ is given by,

$$P_n(a_n) = \frac{|\langle\phi_n|\psi\rangle|^2}{\langle\psi|\psi\rangle}$$

- ✓ The **expectation value** $\langle\hat{A}\rangle$ represents the average result of measuring $\hat{A}$ on the state $|\psi\rangle$. It can be interpreted as an average of a series of measurement of A .

$$\langle\hat{A}\rangle = \frac{\langle\psi|\hat{A}|\psi\rangle}{\langle\psi|\psi\rangle} = \frac{1}{\langle\psi|\psi\rangle} \sum_{n,m} \langle\psi|\phi_m\rangle \langle\phi_m|\hat{A}|\phi_n\rangle \langle\phi_n|\psi\rangle$$

$$\langle\hat{A}\rangle = \sum_n a_n \frac{|\langle\phi_n|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \sum_n a_n P_n$$

Problem Consider a system whose state is given in terms of a complete and orthonormal set of five vectors: $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$, $|\phi_4\rangle$ and $|\phi_5\rangle$ as,

$$|\psi\rangle = \frac{1}{\sqrt{19}}|\phi_1\rangle + \frac{2}{\sqrt{19}}|\phi_2\rangle + \sqrt{\frac{2}{19}}|\phi_3\rangle + \sqrt{\frac{3}{19}}|\phi_4\rangle + \sqrt{\frac{5}{19}}|\phi_5\rangle$$

Here, $|\phi_n\rangle$ are eigenstates of the operator $\hat{H}$ such that $\hat{H}|\phi_n\rangle = n\epsilon_0|\phi_n\rangle$ with $n = 1, 2, 3, 4, 5$ where ϵ_0 has the dimension of energy. If the energy is measured over large number of such identical systems, then find the average energy of one such systems.

Solution: Since, $|\psi\rangle$ is not normalized,

$$\langle\psi|\psi\rangle = \sum_{n=1}^5 c_n^2 \langle\phi_n|\phi_n\rangle = \sum_{n=1}^5 c_n^2 = \frac{1}{19} + \frac{4}{19} + \frac{2}{19} + \frac{3}{19} + \frac{5}{19} = \frac{15}{19}$$

$$P_1(E_1) = \frac{|\langle \phi_1 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{1}{19} \times \frac{19}{15} = \frac{1}{15}$$

$$P_2(E_2) = \frac{|\langle \phi_2 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{4}{15}; \quad P_3(E_3) = \frac{|\langle \phi_3 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{2}{15}$$

$$P_4(E_4) = \frac{|\langle \phi_4 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{3}{15}; \quad P_5(E_5) = \frac{|\langle \phi_5 | \psi \rangle|^2}{\langle \psi | \psi \rangle} = \frac{5}{15}$$

Average energy,

$$E = \langle \hat{H} \rangle = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{19}{15} \sum_{n=1}^5 c_n^2 \langle \phi_n | \hat{H} | \phi_n \rangle$$

$$\Rightarrow E = \frac{19}{15} \left(\frac{1}{19} + \frac{8}{19} + \frac{6}{19} + \frac{12}{19} + \frac{25}{19} \right) \epsilon_0 = \frac{52}{15} \epsilon_0$$

Postulate 5: Time evolution of a system

- ✓ The time evolution of the state vector $|\psi(t)\rangle$ of a system is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H} |\psi(t)\rangle$$

- ✓ Here, $\hat{H}$ is known as the **Hamiltonian operator** which corresponds to the total energy of the system.